Dazaet Galicia-Badillo

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10 October 2026

Dr Daniel Lee
Department of Chemical Engineering, School of Engineering
The University of Manchester
Manchester M13 9PL, United Kingdom

Re: Research Associate in Solid-State NMR Spectroscopy, ref. SAE-032399

Dear Dr Lee,

I am writing to apply for the position of Research Associate in Solid-State NMR Spectroscopy (ref. SAE-032399) within the EPSRC Programme Grant on Molecular Solid Solutions. I completed my PhD in Chemical Sciences at the Institute of Chemistry, UNAM (Mexico), in 2026 under the supervision of Prof. Braulio Rodríguez-Molina, and the degree was awarded with honours.

Solid-state NMR was the central technique of my PhD. On a Bruker 500 MHz spectrometer with a double-resonance (^{1}H/X) MAS probe, I set up my own experiments, including tuning and matching both channels and retuning the ^{1}H channel for ^{19}F. I acquired ^{13}C CP-MAS spectra with variable contact times, ^{19}F MAS spectra, variable-temperature ^{2}H spectra and some ^{15}N CP-MAS spectra. I simulated ^{2}H lineshapes with NMR WebLab to obtain rotational rates and activation energies, and analysed ^{13}C T_1 and ^{1}H and ^{19}F relaxometry data to follow faster motions.

I use NMR together with diffraction. In a pyrrolo[3,2-b]pyrrole solvate that jumps on heating (Mater. Chem. Front., 2026), ^{2}H NMR and variable-temperature single-crystal diffraction allowed me to separate the molecular dynamics from the anisotropic thermal expansion that drives the jump. In a zirconium MOF (ACS Mater. Lett., 2026), the same combination showed how a photoswitch and guest uptake change the motion of a molecular rotor. Both projects were done with Dr Martin Dračínský and Dr Jan Blahut at IOCB Prague.

The role also involves preparing the solid solutions, and synthesis and crystallisation are a large part of my training. I grow single crystals of molecular crystals, cocrystals and solvates and characterise them by SCXRD, PXRD, DSC, TGA and hot-stage microscopy. I contributed to a study in which the salient temperature of cocrystals was tuned through the composition of the crystallisation solvent mixture (Angew. Chem. Int. Ed., 2026). I also run DFT (Gaussian16, periodic CP2K) and molecular dynamics (LAMMPS) on HPC clusters. Diffraction gives the average structure of a molecular solid solution, while solid-state NMR can show how a dopant is distributed and how it moves locally, and this is the question I would like to work on within MoSS.

I have not yet used DNP. The UK DNP MAS NMR Facility at Manchester, and the chance to extend my work to ^{17}O and natural-abundance ^{15}N, are among my main reasons for applying. During my PhD I learned periodic DFT and molecular dynamics from scratch and now use both routinely, and I would approach DNP in the same way.

My 14 publications include collaborations with groups in the Czech Republic, the United States, China and the United Arab Emirates. I co-supervised four bachelor's theses and nine undergraduate research placements, trained students in laboratory procedures and safety, and handled purchasing and inventory for our laboratory. I have presented 15 posters and 9 talks, and received the Best Poster Award at DynaMIC23 in Fribourg. Thank you for considering my application; I would be glad to discuss it at interview.

Yours sincerely,

Dazaet Galicia-Badillo, PhD

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PhD in Chemical Sciences · Solid-state NMR, X-ray diffraction and dynamics of molecular crystals

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Current affiliation: Instituto Mexicano de la Propiedad Industrial (IMPI), Mexico City

PROFILE

Materials chemist trained at the Institute of Chemistry, UNAM, where my PhD used multinuclear solid-state NMR (^{13}C, ^{19}F, ^{2}H, ^{15}N) on a Bruker 500 MHz spectrometer to measure molecular motion in organic crystals and metal-organic frameworks. I analyse NMR data by lineshape simulation and relaxation analysis and interpret it together with variable-temperature single-crystal and powder X-ray diffraction, thermal analysis and DFT. Author of 14 publications, with collaborators in the Czech Republic, the United States, China and the United Arab Emirates.

RESEARCH EXPERIENCE

Doctoral researcher, Institute of Chemistry, UNAM (Prof. Braulio Rodríguez-Molina) 2022 - 2026

- Set up and ran solid-state NMR experiments on a Bruker 500 MHz spectrometer with a double-resonance (^{1}H/X) MAS probe, including probe tuning and matching of both channels (^{1}H channel retuned for ^{19}F): ^{13}C CP-MAS with variable contact times, ^{19}F MAS, variable-temperature ^{2}H and ^{15}N CP-MAS.
- Determined rotational dynamics of mobile components in organic crystals and Zr-based MOFs by ^{2}H lineshape simulation (NMR WebLab) and analysis of ^{13}C T_1 and ^{1}H/^{19}F relaxometry data.
- Correlated NMR dynamics with variable-temperature SCXRD/PXRD structures to separate molecular motion from anisotropic thermal expansion in a thermosalient solvate (*Mater. Chem. Front.* 2026) and to show how a photoswitch and guest uptake modulate a molecular rotor inside a MOF (*ACS Mater. Lett.* 2026), in collaboration with the NMR group of Dr Martin Dračínský (IOCB Prague).
- Characterised molecular crystals, cocrystals and solvates by DSC, TGA and hot-stage microscopy, relating phase transitions, salient behaviour and mechanical flexibility to crystal structure.
- Ran periodic DFT (CP2K) and molecular dynamics (LAMMPS) on HPC clusters; wrote two successful proposals for national supercomputing allocations (LANCAD, 2024).
- Synthesised organic molecular rotors and photoswitches (including Schlenk techniques) and MOFs (solvothermal synthesis and post-synthetic modification); grew single crystals for diffraction.

M.Sc. and undergraduate researcher, Institute of Chemistry, UNAM 2019 - 2022

- Studied guest-dependent rotational dynamics in a zirconia MOF (*Chem. Sci.* 2020) and supramolecular motifs in fluorinated xanthenediols (*ACS Omega* 2022).

TECHNICAL SKILLS

Solid-state NMR: Bruker 500 MHz, TopSpin; double-resonance ^{1}H/X MAS probe tuning and matching; ^{13}C CP-MAS (variable contact time), ^{19}F MAS, variable-temperature ^{2}H, ^{15}N CP-MAS; ^{2}H lineshape simulation (NMR WebLab); ^{13}C T_1 and ^{1}H/^{19}F relaxometry analysis; processing in TopSpin and MestReNova.

Diffraction: SCXRD and PXRD, including variable-temperature studies; structure solution and refinement (SHELX); CCDC suite, CrystalDiffract.

Thermal and optical: DSC, TGA, hot-stage microscopy; UV-Vis absorption; steady-state and time-resolved fluorescence.

Computation: DFT in Gaussian16 (geometry optimisation, rotational energy profiles, PES); periodic DFT in CP2K; molecular dynamics in LAMMPS; HPC job workflows; AI-assisted data analysis.

Synthesis: multistep organic synthesis, air- and moisture-sensitive chemistry, crystallisation of molecular crystals, cocrystals and solvates; MOF synthesis and post-synthetic modification.

Laboratory: training of students in laboratory procedures and safety; purchasing, inventory and organisation of a research laboratory.

EDUCATION

PhD in Chemical Sciences (with honours), UNAM, Institute of Chemistry 2022 - 2026

Thesis: Molecular and Macroscopic Motion in Organic Crystals and Metal-Organic Frameworks. Supervisor: Prof. Braulio Rodríguez-Molina.

M.Sc. in Chemical Sciences, UNAM, Institute of Chemistry 2020 - 2022

B.Sc. in Chemistry, UNAM, Faculty of Chemistry 2015 - 2020

PUBLICATIONS

14 publications: 12 research articles, 1 review and 1 preview (2 in *Matter*).

14. Enriquez-Palacios, E.; **Galicia-Badillo, D.**; Flores-Alamo, M.; Belmonte-Vazquez, J. L. Imidazole-based Photoswitches by Design and their use for Data Encryption. *ACS Appl. Mater. Interfaces* **2026**, 18 (35), 47895-47906.
13. **Galicia-Badillo, D.**; Hernández-Morales, E. A.; López-Herrera, Y.; Hernández-Santiago, E.; Blahut, J.; Toscano, R. A.; Dračínský, M.; Rodríguez-Molina, B. Dissecting Molecular and Macroscopic Dynamics in a Jumping Solvate. *Mater. Chem. Front.* **2026**, 10 (18), 3305-3313.
12. **Galicia-Badillo, D.**; Cisneros, J.; Blahut, J.; Acosta-Vera, A.; García-Márquez, A.; Desirena, H.; Bartůnková, E.; García-González, M. C.; Chávez-Serrano, I.; Rodríguez-Hernández, J.; Dračínský, M.; Rodríguez-Molina, B. Integration of a Molecular Rotor and a Photoswitch in a Metal-Organic Framework. *ACS Mater. Lett.* **2026**, 8 (7), 1995-2001.
11. Hernández-Morales, E. A.; Barreto-Hernández, D.; **Galicia-Badillo, D.**; Toscano, R. A.; García-Aguilera, M. E.; Rodríguez-Molina, B. Crystals as Rockets: Modulation of the Salient Temperature in Cocrystals by Solvent Mixture Composition. *Angew. Chem. Int. Ed.* **2026**, e26010.
10. Álvarez-Castillo, Á. D.; Juárez-Morales, A.; **Galicia-Badillo, D.**; Maldonado-Domínguez, M.; Heredia-Peñaloza, J. M.; Flores-Álamo, M.; Belmonte-Vázquez, J. L. Rational Modulation of Excited-State Intramolecular Proton Transfer, Aggregation-Induced Emission, and Intramolecular Motion on Polysubstituted Imidazoles. *ChemPhotoChem* **2026**, 10, e70099. (Front cover)
9. Castañón-García, M.; López-Mendoza, P.; **Galicia-Badillo, D.**; Miranda, L. D. Synthesis of Non-Symmetrical 3,3'-Bicoumarins: Total Synthesis of Arteminorin C, 3,3'-Biisofraxidin and Biscopoletin. *Org. Lett.* **2025**, 27 (21), 5349-5354.
8. Belmonte-Vázquez, J. L.; Karothu, D. P.; Fleischer, C. H.; **Galicia-Badillo, D.**; Maldonado-Domínguez, M.; Schurko, R. W.; Li, L.; Naumov, P.; Rodríguez-Molina, B. Low-Temperature Dynamics at Nano- and Macroscales: Organic Crystal That Exhibits Low-Temperature Molecular Motion and the Thermosalient Effect. *Chem. Mater.* **2025**, 37, 3373-3383.
7. **Galicia-Badillo, D.**; Rodríguez-Molina, B. On the Flexibility of 1,4-Bis(phenylethynyl)benzene Butyl Ester Organic Crystal Featuring Mobile Components. *Tetrahedron* **2024**, 167, 134245.
6. **Galicia-Badillo, D.**; Belmonte-Vázquez, J. L.; Rodríguez, M.; Rodríguez-Molina, B.; García-González, M. C. Aggregation-Induced Emission Enhancement and Solid-State Photoswitching of Crystalline Carbazole N-Salicylidene Anilines. *ACS Omega* **2024**, 9 (36), 38015-38022.
5. Acosta-Vera, A.; **Galicia-Badillo, D.**; Navarro-Huerta, A.; Stracke, K.; Gómez-Vidales, V.; Rodríguez-Hernández, J.; Evans, J. D.; Rodríguez-Molina, B. Modulation of the Photophysics and Internal Dynamics in a Zr Metal-Organic Framework by the Inclusion of Fluorinated Guests. *ACS Mater. Lett.* **2024**, 6 (9), 4395-4401.
4. **Galicia-Badillo, D.**; García-Vargas, P. N.; Rodríguez-Molina, B. The Integration of Multiple Components for Multi-Responsive Dynamic Framework Materials (Preview). *Matter* **2024**, 7 (8), 2766-2768.
3. Hernández-Santiago, E.; Espinosa-Rocha, J.; **Galicia-Badillo, D.**; Rodríguez-Molina, B. Local and Global Dynamics in Organic Soft Porous Crystals (Review). *Matter* **2024**, 7 (2), 408-429.
2. Rodríguez-Molina, M.; **Galicia-Badillo, D.**; Cetina-Mancilla, E.; Cárdenas, J.; Olvera, L. I.; Toscano, R. A.; Rodríguez-Molina, B.; Zolotukhin, M. G. 9-Trifluoromethylxanthenediols: Synthesis and Supramolecular Motifs. *ACS Omega* **2022**, 7 (16), 13520-13528.
1. Torres-Huerta, A.; **Galicia-Badillo, D.**; Aguilar-Granda, A.; Bryant, J. T.; Uribe-Romo, F. J.; Rodríguez-Molina, B. Multiple Rotational Rates in a Guest-Loaded, Amphidynamic Zirconia Metal-Organic Framework. *Chem. Sci.* **2020**, 11 (42), 11579-11583.

HONOURS, AWARDS AND FUNDING

PhD awarded with honours, UNAM	2026
Nominated for the Alfonso Caso Medal, UNAM (outcome pending)	2026
Best Presentation Award, IX Seminario Regional de Materiales Avanzados, UAEH, Mexico	2025
Two supercomputing allocations, LANCAD (proposals written as lead applicant)	2024
Best Poster Award, DynaMIC23 (Dynamic Materials, Crystals and Phenomena), Fribourg, Switzerland	2023
Doctoral and Master's scholarships, SECIHTI (Mexico)	2020, 2022
Institute of Chemistry Undergraduate Scholarship for best undergraduate project, UNAM	2019

MENTORING AND COLLABORATION

- Co-supervised 4 bachelor's theses (2023 - present) on MOF synthesis, crystal dynamics and luminescent materials; supervised 9 undergraduate research placements (2022 - 2025) and 1 high-school student.
- Contributed experimental work to collaborations with groups in the Czech Republic (IOCB Prague), the United States, China and the United Arab Emirates.

CONFERENCES AND OUTREACH

- 15 posters (6 international) and 9 oral presentations, including 4 invited lectures.
- Outreach: article in Boletín IQ-UNAM (2022), 2 public lectures for high-school audiences (2024 - 2025), media consultation for UNAM Global (2022).

REFEREES

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